

AUTOMATED RUBRIC ASSIGNMENT EVALUATOR

Final Deliverables

Lab 1 — Requirements Engineering & UML Use-Case Modelling

Deliverable	Completed Artifact
RAE-2	System actors
RAE-3	Functional requirements summary
RAE-4	Non-functional requirements / constraints summary
RAE-6	UML use-case diagram
RAE-7	Include and extend relationships
RAE-9	Use-case flow

This document consolidates the completed deliverables available from the project work. Requirement wording is preserved where it is explicitly present in the completed artifacts.

1. System Actors

The Automated Rubric Assignment Evaluator involves the following actors identified for the UML requirement.

Actor	Interaction with the system
Student	Uploads project submissions and views evaluation results.
Faculty Evaluator	Reviews evaluation results, configures/uses evaluation criteria, and can review or request re-evaluation.
System Administrator	Manages users, system configuration, and evaluation infrastructure.

2. Functional Requirements Summary

The exact original FR-001 to FR-005 requirement table was not present in the retrieved completed files. Therefore, the following is a consolidated functional summary based only on behaviors explicitly represented in the completed use-case flow and UML artifacts; it is not presented as a verbatim replacement of the original requirement wording.

ID	Functional behavior evidenced by completed artifacts
FR-001	Generate and display a rubric score breakdown for a submitted project; the completed flow specifies display to the Student.
FR-002	Accept a project ZIP submission and validate its basic structure before automated processing.
FR-003	Queue the submission and execute the configured test suite in an isolated execution environment, returning pass/fail results.
FR-004	Allow the Faculty Evaluator to override an automated score with a justification, validate the adjusted value, and retain the score.
FR-005	Generate, store, and present an evaluation result/report containing the rubric score and evaluation results; support review.

3. Non-Functional Requirements / Quality Constraints

The completed use-case artifact explicitly references NFR-001 and NFR-002. Their original standalone wording was not available in the retrieved files, so the table below records only quality constraints explicitly evidenced by the completed flow.

Evidence	Quality constraint represented in completed artifact
NFR-001 / performance evidence	The rubric breakdown is rendered to the Student and Faculty Evaluator within 10 seconds.
NFR-002 / auditability evidence	Faculty score overrides are timestamped and the original automated score is retained in an audit trail.
Reliability evidence	If evaluation fails, no incomplete score is saved; the submission is flagged for manual review and the user is notified.

4. UML Use-Case Diagram

The completed UML artifact includes the system boundary, Student, Faculty Evaluator, System Administrator, multiple use cases, actor associations, and include/extend relationships.

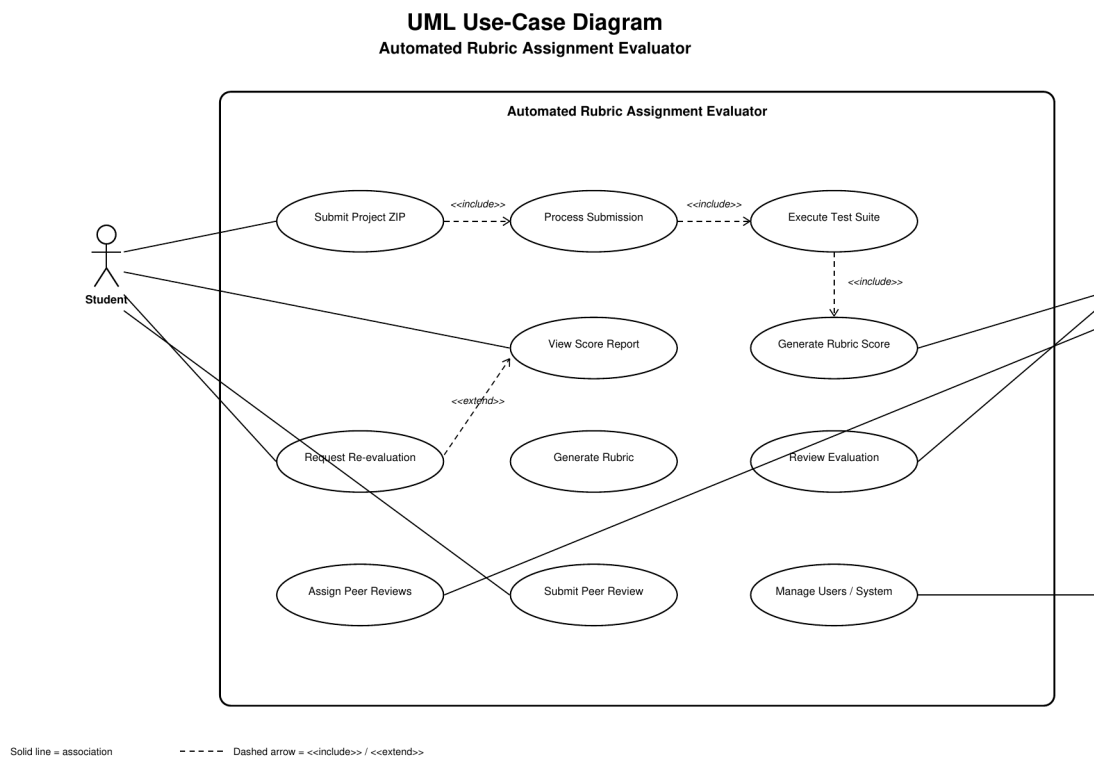


Diagram included from the completed UML deliverable.

5. Include and Extend Relationships

The completed UML diagram represents the following dependency relationships:

Relationship	Purpose represented
<<include>>	Submit Project ZIP includes Process Submission.
<<include>>	Process Submission includes Execute Test Suite.
<<include>>	Execute Test Suite includes Generate Rubric Score.
<<extend>>	Request Re-evaluation extends View Score Report in the completed diagram.

6. Use-Case Flow — Evaluate Project Submission

Primary Actor: Student

Supporting Actors: Faculty Evaluator, System Administrator

Goal: Evaluate a submitted project using the defined rubric and generate an evaluation result.

Preconditions

- The student has a project submission ready.
- The evaluation system and rubric are available.

Postconditions

- The submission is evaluated.
- A rubric score is calculated.
- An evaluation report is generated.

Main Success Scenario

1. Student uploads the project submission in ZIP format.
2. The system receives and validates the submission.
3. The system executes the required test suite on the submission.
4. The system evaluates the submission against the defined rubric.
5. The system calculates the rubric score based on the evaluation results.
6. The system generates an evaluation report containing the results and score.
7. Faculty Evaluator reviews the evaluation result and report.
8. The system stores the evaluation result for future reference.

Alternate / Exception Flow

- A1. If the uploaded ZIP file is invalid or incomplete, the system rejects the submission and asks the Student to upload it again.
- A2. If a test cannot be executed successfully, the system records the failure and includes it in the evaluation result.
- A3. If the Faculty Evaluator identifies an issue with the evaluation, the submission can be reviewed or sent for re-evaluation.
- A4. If re-evaluation is requested, the system performs the evaluation again and generates an updated result.

Relationship with the UML Diagram

The flow corresponds to the main UML use cases: Submit Project (ZIP), Execute Test Suite, Evaluate Submission, Calculate Rubric Score, Generate Evaluation Report, Review Evaluation, and Request Re-evaluation.

7. Final Deliverable Checklist

This combined document brings the completed requirement-analysis and UML artifacts into one submission-ready file.

Item	Included
System actors	✓
Functional requirements summary	✓
Non-functional constraints evidenced by completed artifacts	✓
UML use-case diagram	✓
Include / extend relationships	✓
Use-case flow with preconditions and postconditions	✓
Main success scenario	✓
Alternate / exception flow	✓

Source artifacts consolidated

- UML_Use_Case_Diagram_final.pdf — completed UML artifact.
- RAE-9_Use-Case_Flow.pdf — completed use-case flow artifact.

Note: The final PDF intentionally does not invent missing original requirement wording. The functional and non-functional summaries are based on what is explicitly supported by the completed artifacts.